After the Card: Sickle Cell Post-Screening Care Loop

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Summary

India has screened over 6 crore people for sickle cell disease, but a status card is not care. After screening, patients drop out of follow-up, take hydroxyurea irregularly, miss early crisis signs, and carriers misunderstand what their status means for their children.

We are building a post-screening care loop that turns a screening result into continuous, understood, and acted-upon care, working alongside existing government systems and frontline health workers. Built for UnivaBio 2026 (theme: AI for Human Health).

Problem statement

> India has screened over 6 crore people for sickle cell disease and issued status cards, but a card is not care. After screening, patients drop out of follow-up, hydroxyurea is taken irregularly or never adjusted, warning signs of a crisis are missed at home, and carriers don't understand what their status means for their children. There is no system that turns a screening result into continuous, understood, and acted-upon care in the communities where the disease is most common.

One line: India solved finding sickle cell. Nobody has solved what happens the day after.

Evidence

The research points to three gaps after screening: people leave care, they don't understand their status, and the mission is still measured mainly on screening.

1. Care works, but people fall out of it.

- In a tribal Gujarat program of 404 patients, hydroxyurea coverage rose from 3.5% to 88% and pain crises fell from 277 to 53.4 per 100 person-years. Yet about one quarter were lost to follow-up (Dave et al., 2019).

- Among 31 children in Bhopal, non-adherence to hydroxyurea was linked to drug unavailability (67.7%), improper follow-up (64.5%) and unawareness (54.8%) (JCDR, 2022).
- In an endemic zone, once hydroxyurea is started there is usually no regular follow-up; patients self-maintain and doses are rarely corrected (JPPCH).

2. People hold a card but misunderstand it.

- In Koraput, Odisha, most believed carriers transmit the disease and did not know carrier-carrier marriage is the risk (PMC7475140).
- Of 394 tribal-district adolescents in Maharashtra, only 5% were aware of the disease and none knew how it is transmitted (IJCM, 2024).
- In the Indian SCD Registry, 85.2% of 324 tribal patients did not know their status before enrolling (ISCDR, 2024).

3. The mission's next challenge is after screening. As of 28 July 2025, over 6.04 crore people were screened, 2.16 lakh diagnosed and 16.7 lakh carriers identified (Medical Dialogues). Analysts argue treatment access, counselling outcomes and card use should now be reported, not just screening numbers (IMPRI).

Who it's for

User	What they need after screening
Patients and families (especially parents of children with SCD)	Stay in care, take hydroxyurea regularly, recognise warning signs early
Carriers and young people	Understand what carrier status means for marriage and children
Frontline health workers (ASHAs, health centre staff)	Know who is slipping out of care and who needs a visit, inside work they already do

The system relies on workers already in the mission's workflow, not new dedicated staff.

What already exists and the gap

Existing tools focus on screening and record-keeping; none owns the patient's journey after the card.

Existing tool	What it does	What it leaves open
National sickle cell portal	Digital records of screened people	Day-to-day care and understanding
Indian SCD Registry app	Patient data collection at health centres	Patient- and family-side follow-up
GIPCI counselling card	Colour-coded card explaining status and genetic risk; app proposed	Ongoing treatment and crisis support

Our solution completes these systems rather than competing with them.

Solution

A post-screening care loop that turns a status card into ongoing care. It rests on three pillars, kept open so features can be added later.

1. **Understand.** Make the card meaningful to the person holding it, in their language and at their literacy level, including what carrier status means for marriage and children.
2. **Stay in care.** Keep patients connected to follow-up and treatment, and flag who is slipping away before they are lost.
3. **Act early.** Help families and health workers recognise warning signs and escalate in time.

The loop links the family side and the health worker side, so a missed dose or a warning sign at home becomes a visible action for the worker. The tool supports care and education; it does not diagnose or replace a clinician.

Role of AI

AI is one component of the care loop, not the product. Clear rules and health worker judgement make the care decisions; AI helps people understand and helps workers prioritise.

- **Explaining in plain language:** turning a status card and care guidance into simple, local-language explanations, including by voice.
- **Prioritising follow-up:** highlighting patients who show early signs of dropping out of care, so workers know whom to reach first.

Safety: AI never decides whether a symptom is an emergency. Warning signs follow fixed, guideline-based rules, and anything urgent goes straight to a health worker or facility.

Why it stands out

Judging criterion	How this project answers it
Idea & Innovation	Targets the documented post-screening gap that existing tools leave open; few teams will pick a tribal-health genetic disorder
Implementation	A working loop linking families and health workers, with AI as one part
Health Impact & Rigor	Grounded in peer-reviewed Indian studies and tied to a national mission with a 2047 goal
Design & Usability	Built for low literacy and local languages, with voice support
Presentation	A clear story: 6 crore screened, and what happens the day after

It also covers all three parts of the hackathon theme: detect earlier, understand better, live more easily.

Build approach

- **AI:** Groq API for language and explanation features.
- **Development:** built with Claude Code.
- **Submission:** working prototype, demo video, one-page PDF, and GitHub repo plus code PDF, per UnivaBio requirements.

Open question: which state and language to pilot the demo in.

Sources

- Attrition from care in a tribal SCD cohort, Gujarat (2019)
- Counselling and hydroxyurea compliance, Bhopal (JCDR 2022)

- Real-world HbF status in an endemic zone (JPPCH)
- Premarital counselling awareness, Koraput
- SCD awareness among tribal adolescents, Maharashtra (IJCM 2024)
- Indian SCD Registry findings (2024)
- Over 6 crore screened (Medical Dialogues, Aug 2025)
- Mission analysis (IMPRI)
- Mission overview (UnderstandUPSC)
- GIPCI card (Daily Pioneer)

Secondary sources were used for some figures; confirm against official NSCAEM releases before final submission.